

The Nonlinear Pendulum: Geometry before Formulas

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Idea: Dimensionless pendulum model

For a pendulum of length $L > 0$ in gravitational acceleration $g_0 > 0$,

$$\frac{d^2\theta}{dt^2} + \frac{g_0}{L} \sin \theta = 0.$$

Set

$$\omega_0 = \sqrt{\frac{g_0}{L}}, \quad \tau = \omega_0 t, \quad v = \frac{d\theta}{d\tau}.$$

With dots denoting differentiation with respect to τ , the dimensionless system is

$$\dot{\theta} = v, \quad \dot{v} = -\sin \theta.$$

A dimensionless time interval T corresponds to physical time

$$T \sqrt{\frac{L}{g_0}}.$$

Recall: Circle as an angular state space

The circle is the quotient

$$S^1 = \mathbb{R}/(2\pi\mathbb{Z}).$$

Thus θ and $\theta + 2\pi k$ represent the same angular position for every $k \in \mathbb{Z}$.

Definition: Physical phase space of the pendulum

The **physical phase space** of the pendulum is

$$S^1 \times \mathbb{R},$$

where $\theta \in S^1$ is angular position and $v \in \mathbb{R}$ is angular velocity.

Definition: Universal cover of the pendulum phase space

Define

$$p : \mathbb{R}^2 \rightarrow S^1 \times \mathbb{R}, \quad p(\theta, v) = ([\theta], v),$$

where $[\theta]$ is the equivalence class of θ modulo 2π . The plane $\mathbb{R}^2$, together with the projection p , is the **universal cover** used for the pendulum phase space. In this cover, the angle coordinate is allowed to take every real value.

Visualization: Cylinder and universal cover

The quotient map identifies vertical sides of every strip of width 2π :

$$p(\theta, v) = p(\theta + 2\pi k, v), \quad k \in \mathbb{Z}.$$

$$\mathbb{R}^2 \text{ with } \theta\text{-period } 2\pi$$

$$\downarrow \text{ identify } \theta = -\pi \text{ with } \theta = \pi$$

$$S^1 \times \mathbb{R}$$

Definition: Contractible closed curve

A closed curve in $S^1 \times \mathbb{R}$ is **contractible** if it can be continuously deformed to a point within $S^1 \times \mathbb{R}$. Otherwise it is **noncontractible**.

Question: Which physical equilibria does the pendulum have?

Find all equilibria of the following system and determine what linearization says about each one:

$$\dot{\theta} = v, \quad \dot{v} = -\sin \theta.$$

Solution. The equilibrium equations give

$$(\theta_*, v_*) = (k\pi, 0), \quad k \in \mathbb{Z}.$$

Modulo 2π , the two equilibria are $(0, 0)$ and $(\pi, 0)$. Since

$$DF(\theta, v) = \begin{pmatrix} 0 & 1 \\ -\cos \theta & 0 \end{pmatrix},$$

we have

$$DF(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \lambda = \pm i,$$

and

$$DF(\pi, 0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \pm 1.$$

Question: Which physical equilibria does the pendulum have? (continued)

Thus linearization is inconclusive at $(0, 0)$, while $(\pi, 0)$ is a hyperbolic saddle. Its stable and unstable eigenspaces are

$$E^s = \text{span}\{(1, -1)\}, \quad E^u = \text{span}\{(1, 1)\}.$$

Theorem: Conservation of pendulum energy

Every solution of

$$\dot{\theta} = v, \quad \dot{v} = -\sin \theta$$

conserves

$$E(\theta, v) = \frac{1}{2}v^2 - \cos \theta.$$

Proof. Along a solution,

$$\dot{E} = v\dot{v} + \sin \theta \dot{\theta} = v(-\sin \theta) + \sin \theta v = 0.$$

Therefore $E(\theta(\tau), v(\tau))$ is constant. □

Definition: Lift of a pendulum motion

Let $z : \mathbb{R} \rightarrow S^1 \times \mathbb{R}$ be a pendulum motion. A **lift** of z is a continuous curve

$$\tilde{z} = (\tilde{\theta}, v) : \mathbb{R} \rightarrow \mathbb{R}^2$$

such that

$$p(\tilde{z}(\tau)) = z(\tau) \quad (\tau \in \mathbb{R}).$$

Definition: Libration

A pendulum motion is a **libration** if it has a lift $\tilde{z} = (\tilde{\theta}, v)$ for which some $T > 0$ satisfies

$$\tilde{\theta}(\tau + T) = \tilde{\theta}(\tau) \quad (\tau \in \mathbb{R}),$$

and $\tilde{\theta}$ oscillates between two turning angles whose difference is less than 2π .

Definition: Rotation

A pendulum motion is a **rotation** if it has a lift satisfying

$$\tilde{\theta}(\tau + T) = \tilde{\theta}(\tau) + 2\pi m$$

for every $\tau \in \mathbb{R}$, for some $T > 0$ and $m \in \{-1, 1\}$.

Recall: Separatrix

A **separatrix** is an invariant trajectory that separates regions of the phase space having qualitatively different motions.

Visualization: Pendulum energy levels in the universal cover

For every $k \in \mathbb{Z}$, the energy levels repeat with period 2π in θ . On the energy level $E = h$,

$$v = \pm \sqrt{2(h + \cos \theta)}.$$

h	level-set geometry	motion
-1	$(2k\pi, 0)$	hanging equilibrium
$-1 < h < 1$	closed oval about $(2k\pi, 0)$	libration
1	saddles and their connecting separatrices	separatrix motion
$h > 1$	one graph above and one below the axis	rotation

Visualization: The separatrix

At $E = 1$,

$$v^2 = 2(1 + \cos \theta) = 4 \cos^2 \left(\frac{\theta}{2} \right).$$

For $-\pi \leq \theta \leq \pi$,

$$v = 2 \cos \left(\frac{\theta}{2} \right)$$

runs from $(-\pi, 0)$ to $(\pi, 0)$ as time increases. It approaches $(-\pi, 0)$ as $\tau \rightarrow -\infty$ and $(\pi, 0)$ as $\tau \rightarrow +\infty$. The lower branch

$$v = -2 \cos \left(\frac{\theta}{2} \right)$$

runs from $(\pi, 0)$ to $(-\pi, 0)$ as time increases, with the opposite limits as $\tau \rightarrow \pm\infty$.

Proposition: The hanging equilibrium is a nonlinear center

The equilibrium $(0, 0)$ of the undamped pendulum is a nonlinear center.

Proof. For each $-1 < h < 1$, the component of $E = h$ surrounding $(0, 0)$ is

$$v = \pm \sqrt{2(h + \cos \theta)}, \quad |\theta| \leq \arccos(-h).$$

It is a compact simple closed curve containing no equilibrium. The vector field is nonzero on the curve, so each trajectory on it is periodic. These curves surround $(0, 0)$ for every h sufficiently close to -1 . $\square$

Visualization: Pendulum portrait on the cylinder

Under the projection $p(\theta, v) = ([\theta], v)$, for $k \in \mathbb{Z}$,

universal cover $\mathbb{R}^2$	cylinder $S^1 \times \mathbb{R}$
libration oval	contractible closed orbit
rotation graph	noncontractible closed orbit
heteroclinic saddle connection	homoclinic orbit
$(2k\pi, 0)$	one hanging equilibrium
$((2k+1)\pi, 0)$	one inverted equilibrium

Observation: Energy threshold

For the undamped pendulum, the energy level $E = 1$ separates libration from rotation.

Question: How fast must the pendulum start to rotate?

Suppose

$$\theta(0) = 0, \quad v(0) = v_0.$$

Classify the motion from v_0 .

Solution. The conserved energy is

$$h = E(0, v_0) = \frac{1}{2}v_0^2 - 1.$$

Therefore

$ v_0 $	h	motion
0	-1	equilibrium
$0 < v_0 < 2$	$-1 < h < 1$	libration
$ v_0 = 2$	$h = 1$	separatrix motion
$ v_0 > 2$	$h > 1$	rotation

For $v_0 > 0$ the angle initially increases; for $v_0 < 0$ it initially decreases.

Proposition: Energy dissipation in the damped pendulum

Let $b > 0$. Every solution of

$$\ddot{\theta} + b\dot{\theta} + \sin \theta = 0$$

satisfies

$$\dot{E} = -bv^2 \leq 0, \quad E(\theta, v) = \frac{1}{2}v^2 - \cos \theta, \quad v = \dot{\theta}.$$

Proof. The first-order system is

$$\dot{\theta} = v, \quad \dot{v} = -bv - \sin \theta.$$

Hence

$$\dot{E} = v(\dot{v} + \sin \theta) = -bv^2.$$

□

Theorem: Damping forbids nonconstant periodic motion

If $b > 0$, every periodic solution of

$$\ddot{\theta} + b\dot{\theta} + \sin \theta = 0$$

is an equilibrium.

Proof. Set

$$v = \dot{\theta}, \quad E(\theta, v) = \frac{1}{2}v^2 - \cos \theta.$$

If a solution has period $T > 0$, then

$$0 = E(\theta(T), v(T)) - E(\theta(0), v(0)) = -b \int_0^T v(\tau)^2 d\tau.$$

Thus $v \equiv 0$. The equation then gives $\sin \theta \equiv 0$, so the solution is an equilibrium. $\square$

Visualization: Damped equilibrium types

For the damped system

$$\dot{\theta} = v, \quad \dot{v} = -bv - \sin \theta, \quad b > 0,$$

the characteristic polynomial at the hanging equilibrium is

$$\lambda^2 + b\lambda + 1.$$

b	local type
$0 < b < 2$	asymptotically stable spiral
$b = 2$	asymptotically stable degenerate node
$b > 2$	asymptotically stable node

At the inverted equilibrium, the Jacobian has determinant $-1 < 0$, so the equilibrium is a saddle for every $b > 0$.

Observation: Direction of energy flow

For the damped pendulum

$$\dot{\theta} = v, \quad \dot{v} = -bv - \sin \theta, \quad b > 0,$$

with energy

$$E(\theta, v) = \frac{1}{2}v^2 - \cos \theta,$$

every solution satisfies

$$\dot{E} = -bv^2.$$

Therefore, away from $v = 0$,

$$\dot{E} = -bv^2 < 0.$$

Thus damped trajectories cross the undamped energy contours toward smaller values of E .